



Stimuli-responsive composite coating based on dynamic metal-ligand interactions for sustainable marine antifouling

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ABSTRACT

While copper-based antifoulants represent effective solutions for marine biofouling prevention, their utilization efficiency remains a critical challenge. This investigation demonstrates a pH-responsive hydrogel composite through synergistic integration of poly(acrylic acid) networks with histidine-coordinated cupric nanoclusters. The main ligand configuration of histidine and Cu²⁺ was identified as “imidazole-dominated and carboxyl-assisted” by density functional theory (DFT) simulation, which provides theoretical support for the pH sensitivity of hydrogel. This coating releases Cu²⁺ precisely “on demand” in response to external pH changes, effectively preventing algae and fouling organisms from adhering and achieving high bactericidal rates against *Pseudomonas aeruginosa* and *E. coli*. The distinctive molecular design of the coating combines multiple amide motifs that self-assemble into a honeycomb network via strong hydrogen bonding, while the coordination between Cu²⁺ and histidine establishes a dynamic metal-ligand interaction with tunable dissociation kinetics and cohesion strength. This intelligent controlled-release composite hydrogel significantly reduces the risk of marine fouling by releasing antifouling agents precisely and efficiently, providing an innovative solution for building a sustainable marine antifouling system.

1. Introduction

Marine biofouling is a serious and long-standing challenge in the field of marine engineering. Marine fouling refers mainly to the process of attachment, colonization and aggregation of marine organisms on solid surfaces, such as the attachment of shellfish, algae and microorganisms [1,2]. Studies by the International Maritime Organization (IMO) have shown that the increase in fuel consumption due to biofouling of ships can be as high as 40 %, costing the global shipping industry more than 150 billion dollars annually [3,4]. In addition, biofouling may trigger species invasions and disrupt regional ecological balance [5]. Among the numerous marine metallic materials, titanium alloys are the most severely subject to biofouling. The surface free energy of titanium alloys (~45 mJ/m²) is in the critical range where organisms are most likely to attach, and the nano passivated film formed on the surface of titanium alloys provides an ideal interface for colonization by organisms [6–8]. The acidic environment generated by the attachment of fouling organisms can cause severe microbiological

influenced corrosion (MIC), which not only challenges the long-term performance as well as puts new demands on the surface modification and protection technologies of titanium alloys.

Traditional antifouling technology relies mainly on biotoxic antifouling agents, and organotin compounds were considered the most effective antifouling materials in the early years [9]. However, the ban on these substances by the IMO underscores the urgent need for a paradigm shift towards sustainable marine protection technologies. In recent years, significant progress has been made in the development of sustainable marine antifouling coatings, such as silicone coatings with low surface energy properties [10–13], biomimetic coatings [14] and zwitterionic polymer coatings [15]. To enhance antifouling performance, composite coatings are usually engineered through the incorporation of highly effective biocides, particularly copper-based antifoulants. Nevertheless, conventional copper-based antifouling systems present their own sustainability challenges. A major drawback is the uncontrolled release kinetics of Cu²⁺, often characterized by an “initial burst and final decay” [16]. This phenomenon is indicative of

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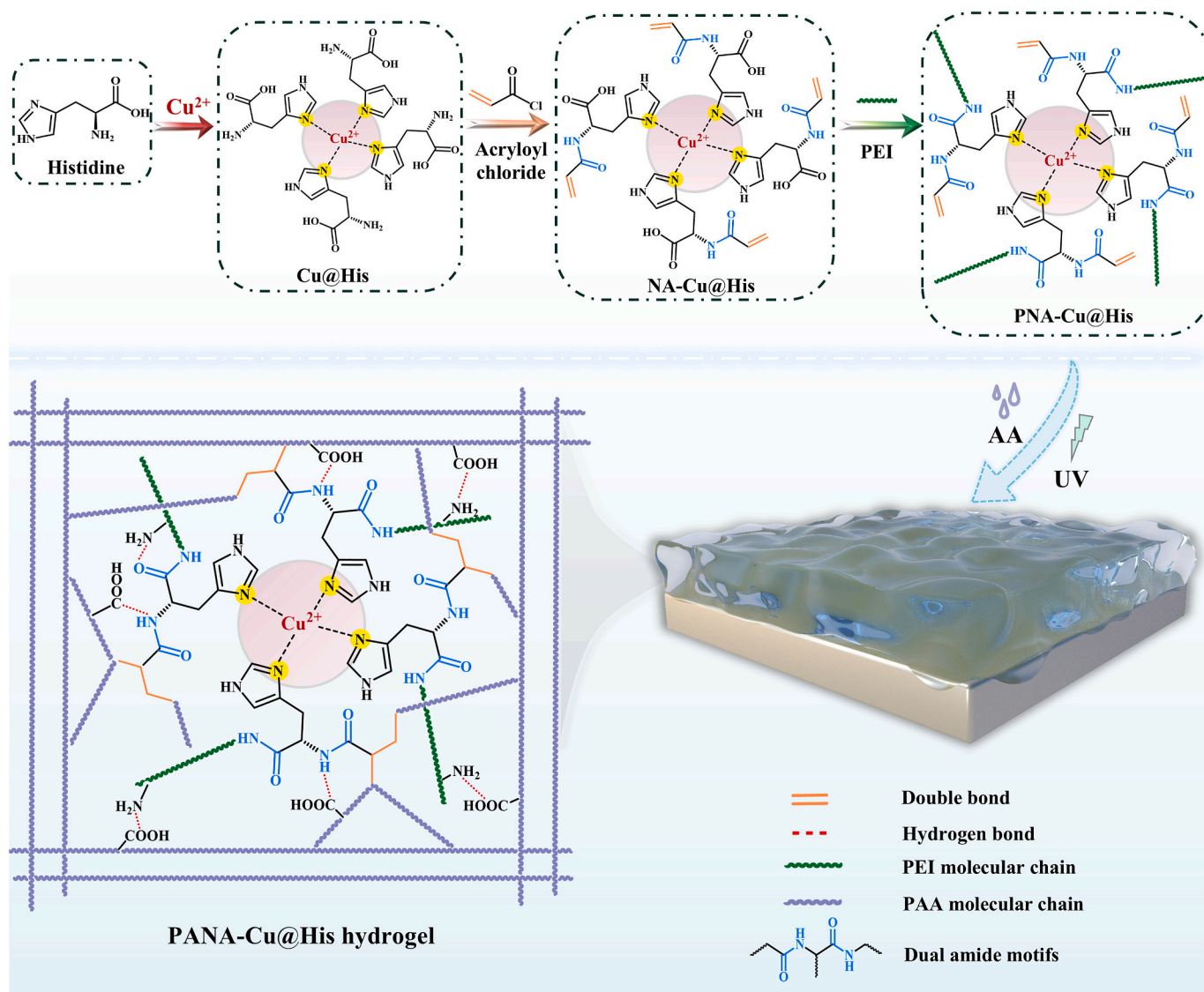
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Scheme 1. Preparation process and molecular structure of PANA-Cu@His hydrogel.

inefficient resource utilization, with a significant amount of active agent being wasted early in the life cycle of the coatings. Moreover, this initial high release of copper can pose risks of acute, localized ecotoxicity to non-target marine life. As the release rate decreases over time, the protective function of the coating diminishes, thus affecting its long-term efficacy and durability. Optimizing the release kinetics of copper ions is critical to advancing copper-based materials towards truly sustainable and effective antifouling solutions.

Hydrogel materials are considered as potential candidates in the field of antifouling due to their excellent hydrophilicity and unique three-dimensional structure [17]. The hydrogel coating locks a “water barrier” on its surface when immersed in solution, effectively blocking the attachment of organic matter, microorganisms and marine organisms [18–20]. However, the harsh marine environment and the diversity of fouling organisms put forward higher requirements for hydrogel coatings. Improving the mechanical, adhesion and anti-fouling properties of hydrogels has become the focus of current research. Dai et al. [21] reported a high-strength hydrogel with dual amide motifs inspired by hydrogen-bonding clusters in protein, demonstrating that the dual amide motifs can form a highly stable hydrogen-bonding structure, thereby enhancing the mechanical properties of the hydrogel. In order to improve the adhesion of the hydrogel coating to the substrate, Li et al.

[22] designed a joint molecule with a carboxyl group to achieve a high adhesion energy ($> 1000 \text{ J/m}^2$) between the hydrogel and the metal. Unfortunately, although carboxylic acid groups can form strong interaction forces with metals and metal oxides, the swelling phenomenon of hydrogels in water still weakens the adhesion of coatings. The development of a hydrogel coating with adjustable antimicrobial properties, high adhesion and excellent mechanical properties is an urgent need in the field of marine antifouling.

As an imidazole-containing basic amino acid with excellent coordination ability with a variety of transition metal ions, histidine is widely used in biomedicine [23], environmental remediation [24] and biomimetic catalysis [25]. Histidine with multiple pH-sensitive moieties can be used to design intelligent stimuli-responsive materials [26], whose potential applications in marine antifouling strategies have not been fully exploited. Based on the above studies, a highly crosslinked hydrogel with acid-sensitive controlled release of Cu^{2+} antifouling agent was explored in this work (Scheme 1). The hydrogel coating consists of polyacrylic acid as matrix, acryloyl chloride and polyethyleneimine as modifiers, and histidine-encapsulated Cu^{2+} as antifouling agent. Specifically, the abundance of carboxylic acid groups within polyacrylic acid facilitates robust chelation of the hydrogel to titanium alloy surfaces. Modification with acryloyl chloride and polyethyleneimine not

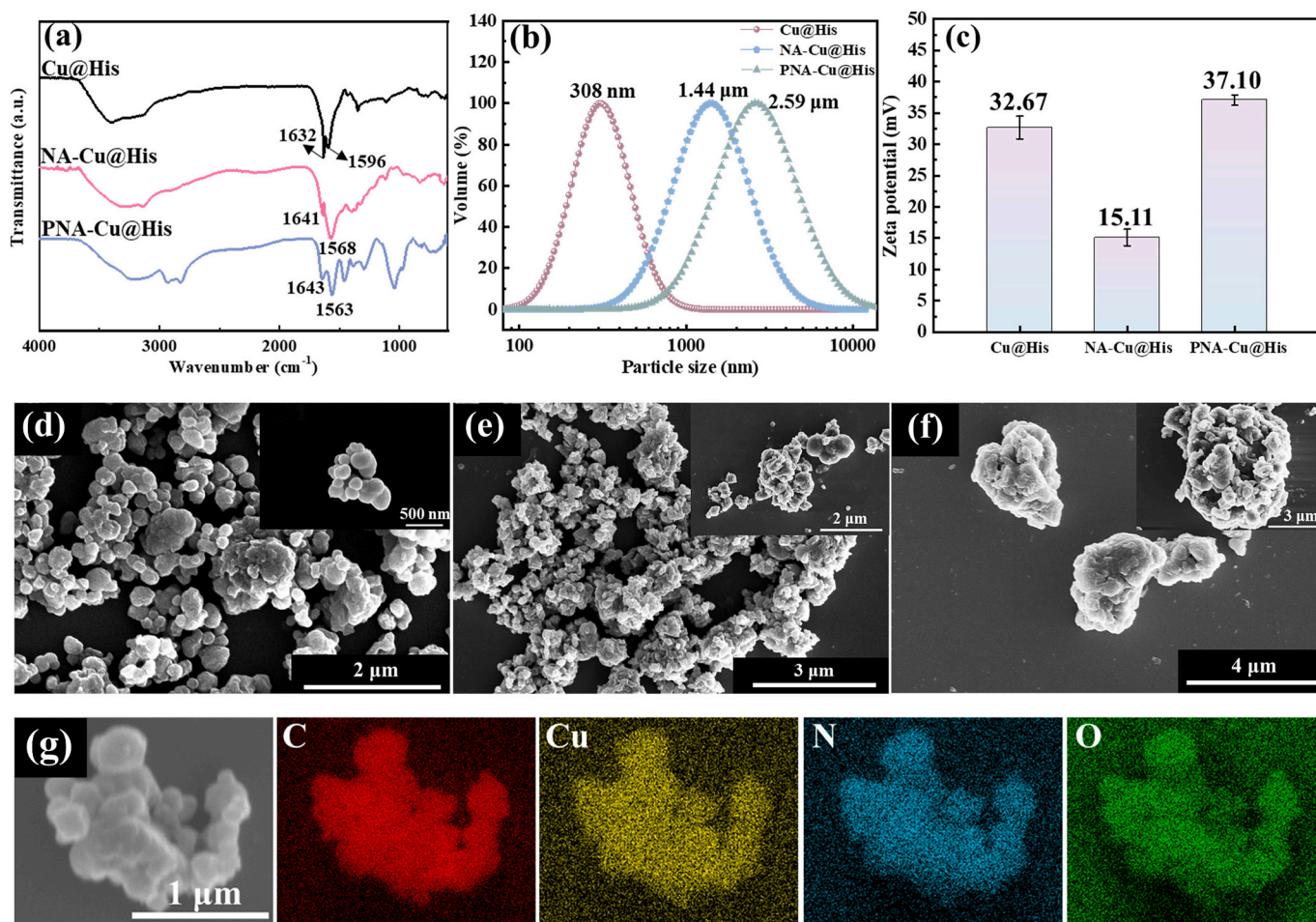


Fig. 1. FTIR spectra (a), particle size distributions (b) and zeta potentials (c) of the prepared particles; SEM images of Cu@His (d), NA-Cu@His (e) and PNA-Cu@His (f); Element mapping of Cu@His (g).

only forms multi-amide motifs in the molecular chain but also introduces reactive double bonds and amino groups, facilitating the formation of a highly cross-linked structure. In contrast to traditional antifouling systems, the coating facilitates a pH-responsive smart-release mechanism for Cu²⁺, minimizing the premature or non-targeted release of antifouling agents, thereby enhancing material efficiency. This antifouling system achieves the dual objectives of effective marine antifouling and minimized environmental impact, representing a sustainable and eco-conscious approach.

2. Experimental procedures

2.1. Reagents and materials

Copper chloride dihydrate (CuCl₂·2H₂O, AR), tetrahydrofuran (99.5 %), acryloyl chloride (96 %), polyethyleneimine (M.W. 10,000), 1-(3-Dimethylaminopropyl)-3-ethylcarbodiimide (EDC, 95 %), N-Hydroxysuccinimide (NHS, 98 %), Acrylic acid (AA, 96 %), ethylene glycol dimethacrylate (crosslinker, 98 %), sodium chloride (AR), 2-Hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone (photo initiator, 98 %), hydrochloric acid (36 % ~ 38 %), nitric acid (65 % ~ 68 %), Cu solution (1000 μg/mL) were received from Shanghai Macklin Reagent Co., Ltd. Ethyl acetate (99.5 %), L-histidine (99 %), potassium carbonate (AR, 99 %), Luria-Bertani medium were obtained from Shanghai Aladdin Biochemical Technology Co., Ltd. Titanium alloys were provided by TANKO Co., Ltd. Artificial seawater (ASW) was prepared in accordance with GB/T 7790-1996. All reagents are not further purified.

2.2. Synthesis of the antifouling hydrogels

2.2.1. Cu@His particles

Cu@His was prepared according to reference with appropriate modifications [27]. 0.425 g of CuCl₂·2H₂O (2.5 mM) was dissolved in 80 mL of deionized water, and 3.875 g of L-histidine (25 mM) was added to the mixed solution with a quick stir. The dark blue mixture was transferred to a reaction kettle and placed in an oven at 180 °C for 12 h. The obtained reddish-brown Cu@His powder was washed three times with deionized water and dried in a vacuum oven at 40 °C.

2.2.2. NA-Cu@His

0.1 g Cu@His and 3 mL of 2 M Na₂CO₃ solution were added to a round bottom flask and stirred for 15 min in ice bath. After the powder was completely dissolved, 1 mL of tetrahydrofuran was added and stirring was continued for 5 min. Subsequently, 110 μL of acryloyl chloride dissolved in 1 mL of tetrahydrofuran was added dropwise to the mixed solution at 5 °C. The mixture was reacted for 4 h at room temperature and the organic phase was removed from the solution with ethyl acetate. The aqueous phase was centrifuged to obtain the black N-acryloyl-Cu@His (NA-Cu@His) particles and washed thoroughly with deionized water.

2.2.3. PNA-Cu@His

0.08 g NA-Cu@His was dissolved in 10 mL of phosphate buffer (pH = 5.8), then 0.1 g EDC and 0.08 g NHS were added to the mixture. After 15 min, 0.8 g of PEI was mixed and stirred for 24 h. The mixture was transferred to a dialysis bag and dialyzed in deionized water for 7 days to

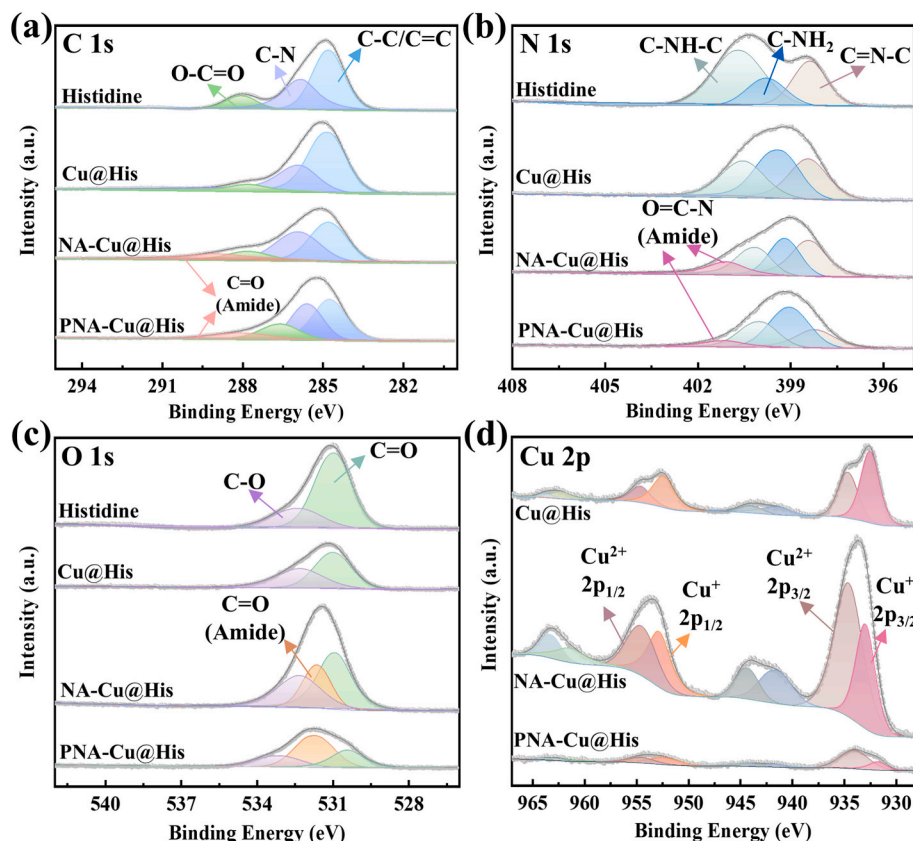


Fig. 2. XPS spectra of C 1s (a), N 1s (b), O 1s (c) and Cu 2p (d) for prepared particles.

remove impurities. The purified samples were freeze-dried and named PNA-Cu@His.

2.2.4. Hydrogel coatings

The monomer, initiator, crosslinker, and water comprise the hydrogel precursor, with the initiator and crosslinker accounting for 1 wt% of the monomer, respectively. The precursor liquid was dropped into the mold and cured under UV light at a wavelength of 365 nm for 30 min to obtain the hydrogels. The hydrogel samples were labeled PANA-Cu@His-0.4 wt%, PANA-Cu@His-0.7 wt%, and PANA-Cu@His-1.0 wt% based on PNA-Cu@His concentration (Relative to the mass of PAA). Pure PAA hydrogels, PAA/PEI hydrogels (the proportion of PEI was 0.7 wt% of PAA) and PAA-Cu@His (1 wt% for Cu@His particles) were also prepared according to the method described above as controls possessing different cross-linking types.

2.3. Characterizations

The morphology images of particles and hydrogels obtained by scanning electron microscopy (Thermoscientific Apreo 2S HiVac). Fourier transform infrared and X-ray photoelectron spectra of particles detected by PerkinElmer Spectrum 100 series spectrometer and X-ray photoelectron spectrometer (Thermo Fisher, ESCALAB QXi). The particle size and zeta potential of the particulate samples were tested by a nanoparticle size and zeta potential analyzer (Brookhaven Instruments Corporation, NanoBrook Omni).

2.4. Mechanical properties of hydrogels

2.4.1. Tensile test

The samples were made in the shape of standard dumbbells (75 × 4 × 2 mm) for tensile testing and the results were obtained by a universal material tester (Trapeziumx-V, AGX-V. 50 N). The number of samples

tested in each group was not less than five.

2.4.2. Swelling behavior

The hydrogel samples (20 × 20 × 2 mm) were dried in an oven to a constant weight and recorded as W_0 . Subsequently, the samples were immersed in ASW for a period of time and then weighed by drying the surface water and recorded as W_t . The swelling rate is calculated by the following formula:

$$\text{Swelling rate (\%)} = \frac{W_t - W_0}{W_0} \quad (1)$$

Each group contains 3 identical samples and the results are averaged over the 3 samples.

2.4.3. Adhesion test

The adhesion strength of the hydrogels on the surface of titanium alloys according to criterion ASTM D4541-09 was obtained from the pull-out test [28]. A 20 mm diameter pull-out nail was glued to the surface of the coating (1.8–2.0 mm), and the pull-out tests were carried out with an adhesion tester of model PosiTest AT-M. The thickness of the coating was 1.8 to 2.0 mm, and the average value was taken from three tests for each sample.

In order to simulate the stability of hydrogel coatings in real marine environments, 10 days of real seawater immersion experiments were carried out and the flow of seawater was simulated with stirring paddles (600 rpm), the seawater was taken from the South China Sea.

2.5. Evaluation of antifouling properties

2.5.1. Antibacterial property

In general, the plate colony counting method provides a visual assessment on the bactericidal capacity of coatings. Titanium alloy components in seawater are mainly subjected to microbial corrosion

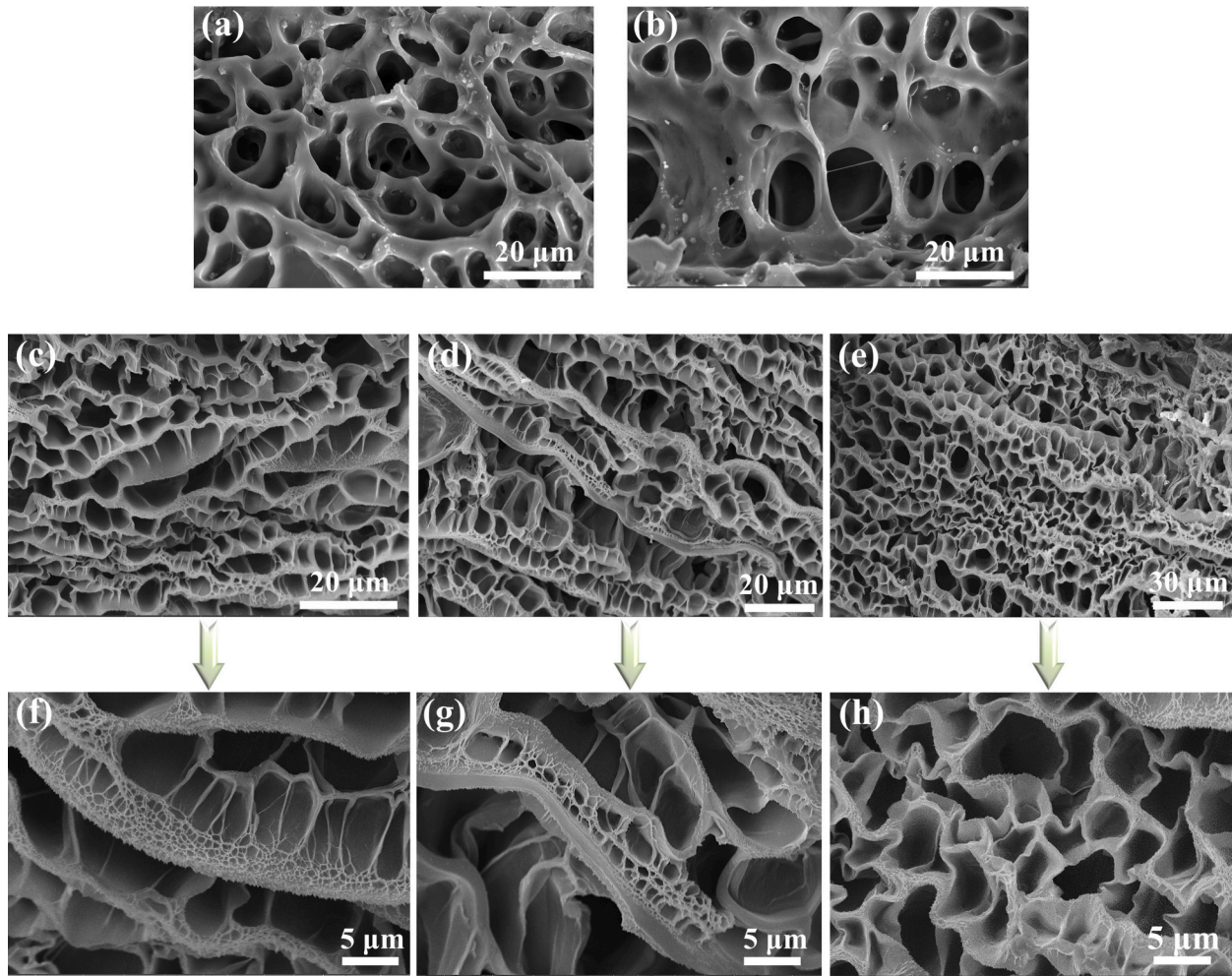


Fig. 3. SEM images of PAA (a), PAA/PEI (b), PANA-Cu@His-0.4 wt% (c, f), PANA-Cu@His-0.7 wt% (d, g), and PANA-Cu@His-1.0 wt% (e, h).

caused by gram-negative bacteria, so the most representative strains of *Pseudomonas aeruginosa* (*P. sa*) and *Escherichia coli* (*E. coli*) were selected for the antibacterial experiment. The strains were added to LB liquid medium and incubated in a constant temperature shaker (37 °C) for 6 h. The cultured bacterial solution was diluted to 10^5 CFU/mL with sterilized ASW. The samples were separately immersed in 30 mL of diluted bacterial solution and continued to be incubated for 24 h. After incubation, the bacterial solution continued to be diluted to 10^2 CFU/mL with ASW. 100 μ L of the diluted bacterial solution was evenly spread on LB agar plates and placed in an incubator at 37 °C for 12 h. The sterilization rates of the coatings are given by the following equation:

$$\text{Sterilization rate (\%)} = \frac{N_c - N_s}{N_c} \quad (2)$$

N_c and N_s are the number of plate colonies in the control and sample groups, respectively. The bacterial solutions soaked with different hydrogels for 24 h were stained and observed under a super-resolution fluorescence microscope (Nikon Corporation).

2.5.2. Anti-algal adhesion property

Chlorella was used to evaluate the algal adhesion resistance of each sample. Typically, the samples were immersed in a certain concentration of *Chlorella* solution and incubated for 14 days. The immersed samples were gently washed with deionized water to remove the poorly adhered *Chlorella*, followed by a 2.5 % glutaraldehyde fixative for 2 h. The processed samples were observed by confocal laser scanning microscope (Leica, Stellaris 5). The above samples were immersed in ethanol

solutions with different concentration gradients, and the dehydrated samples obtained were observed by scanning electron microscopy.

2.5.3. Anti-marine organism adhesion property

Live conchs adhering to the beach reef were collected after the sea water receded and experiments were carried out immediately to ensure the activity of conchs. Live conchs were placed on different samples for three days in seawater to observe the behavior of the conchs, and inversion experiments were used to confirm the adhesion of conchs on various surfaces.

2.6. Measurement of copper ion release rate

The concentration of copper ions released from hydrogels in seawater was determined by atomic absorption spectrometry [29]. Briefly, the PANA-Cu@His-1.0 wt% samples were individually immersed in 50 mL of ASW at different pH with constant stirring. The seawater obtained after a period of immersion was diluted with 0.5 wt% nitric acid and the corresponding copper ion concentrations were measured using an atomic absorption spectrometer (Thermo, SOLAAR ICE 350). The average release rate of Cu^{2+} was calculated by the following equation:

$$R_{\text{release}} = \frac{C_{\text{ion}} \times V_{\text{ASW}}}{S_{\text{coating}} \times t} \quad (3)$$

where C_{ion} is the ion concentration ($\mu\text{g/L}$), V_{ASW} is the solution volume (L), S_{coating} is the specimen area (cm^2), and t is the immersion time (day).

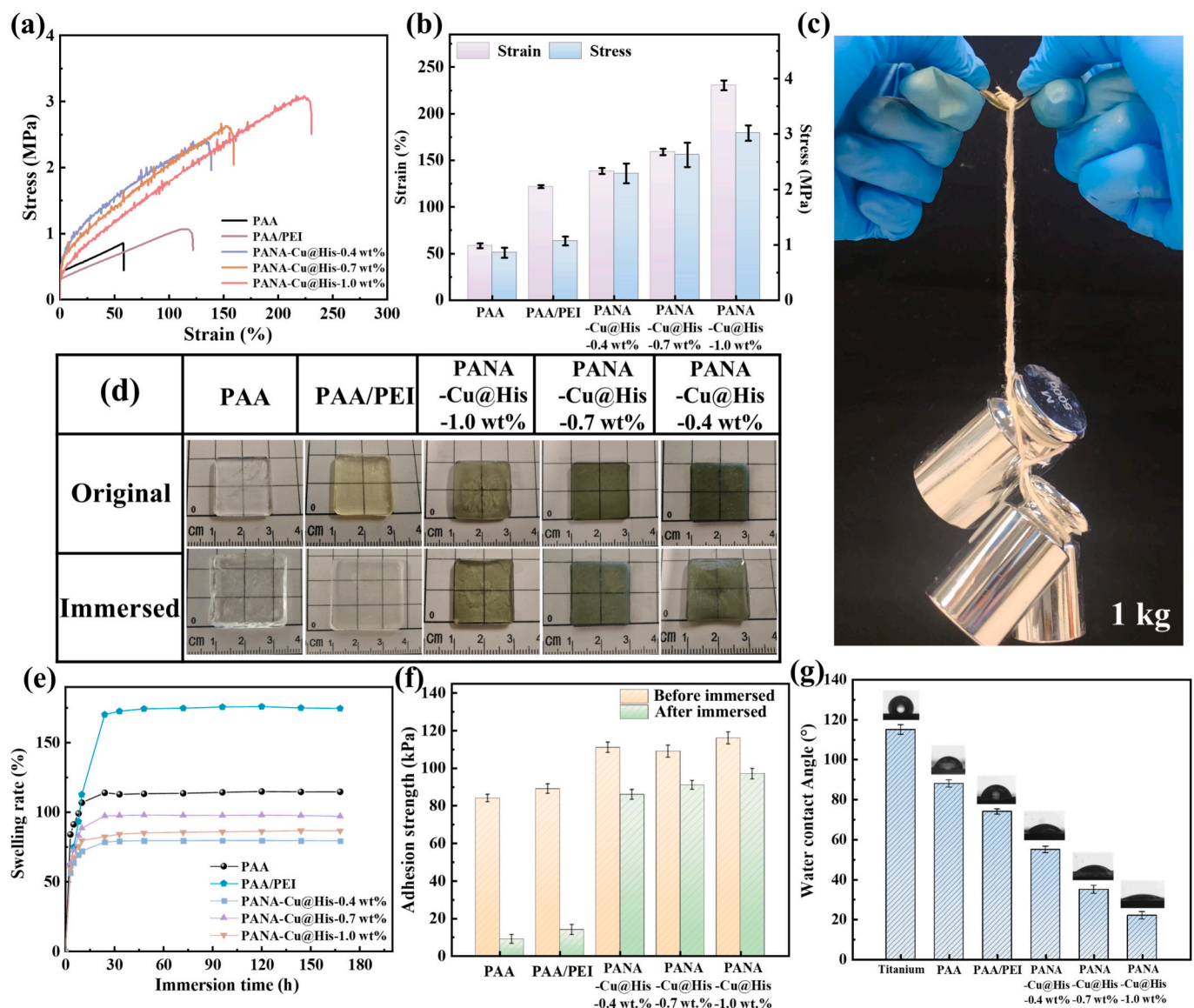


Fig. 4. Tensile curves (a), stress and strain values (b), load-bearing test (c), swelling photographs (d), swelling rate curves (e) adhesion strength (f) and water contact angle (g) of hydrogel samples.

2.7. Details of DFT calculations

Optimal interaction sites between copper ions and histidine were explored using Gaussian 16 software. All structure optimization and binding energy calculations were performed by the DFT-B3LYP method with 6-311++G (d, p) basis set [30]. The inherent limitations in conventional DFT frameworks were addressed through incorporation of the DFT-D3 empirical dispersion scheme [31]. Considering the solvation effect of the system, all calculations were based on the SMD solvent model. To elucidate the nucleophilic and electrophilic regions of histidine, electrostatic potential maps were computationally derived employing the Multiwfn and VMD software suites [32,33]. The binding energy can be calculated by the following formula:

$$E_{\text{Binding}} = E_{\text{Cu@His}} - E_{\text{Cu}^{2+}} - E_{\text{His}} \quad (4)$$

3. Results and discussion

3.1. Structural characterization of particles and hydrogels

The FTIR spectra located in Fig. 1a initially identified the chemical

structure of the prepared Cu@His, NA-Cu@His, and PNA-Cu@His particles. Broad peaks located at 3100–3500 cm^{-1} were observed in Cu@His, where the high and low bands corresponded to the N–H symmetric stretching in the amino and imidazolyl groups, respectively [34]. In addition, a stretching vibration peak of the deprotonated carboxyl group appeared at 1632 cm^{-1} and the peak at 1596 cm^{-1} corresponded to the stretching vibration of the imidazole group in histidine. Notably, the deprotonated carboxyl peak disappeared after subsequent reactions, suggesting that the carboxyl group of Cu@His is free and that the carboxyl group may not be the main binding site for Cu^{2+} , as confirmed by DFT calculations [35]. In NA-Cu@His, the peaks at 3400 cm^{-1} were smoothed and the appearance of peaks at 1641 cm^{-1} (C=C), 1568 cm^{-1} (amide N–H) indicated the successful modification of histidine by acryloyl chloride. The classical C–H stretching peaks of PEI appeared in PNA-Cu@His at 2931 cm^{-1} and 2822 cm^{-1} [36]. The particle size distributions of the particles are shown in Fig. 1b. With the modification of acryloyl chloride, the particle size of the particles increased from 308 nm to 1.44 μm , which was attributed to the formation of amide bonds that increased the inter-particle attraction. The further larger particle size (2.59 μm) of PNA-Cu@His indicates the

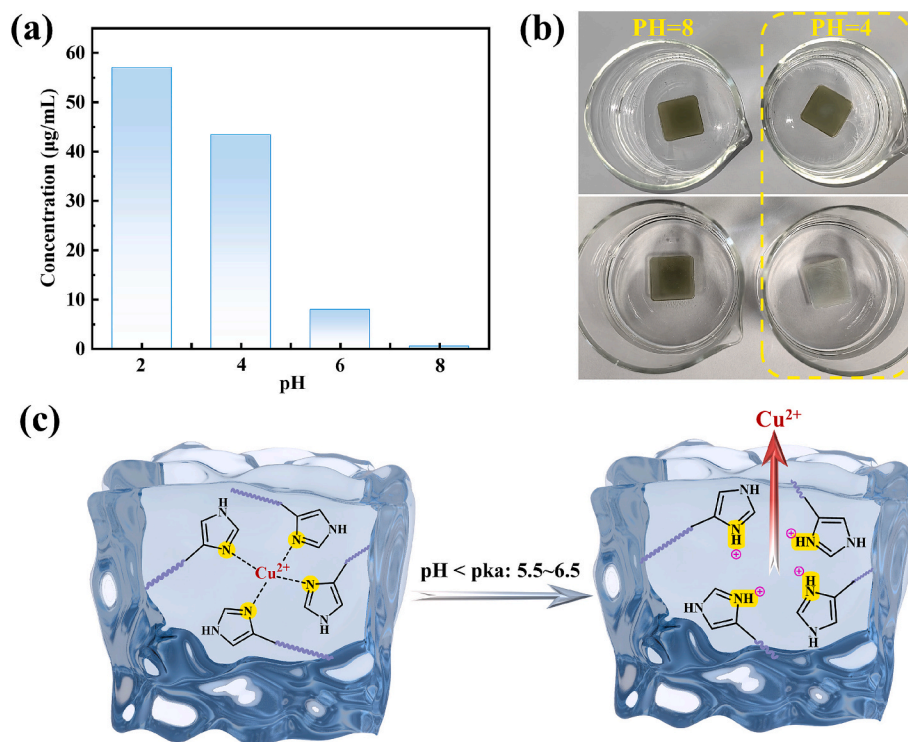


Fig. 5. Concentration of metal ions released (a) and changes in the appearance (b) from PANA-Cu@His-1.0 wt% immersed in seawater of different pH; Mechanism of acid-sensitive Cu^{2+} release from hydrogels (c).

successful loading of PEI. Fig. 1c illustrates the zeta potential of the samples. The Cu@His particles exhibit a positive charge of 32.67 mV due to the presence of amino groups and copper ions. Interestingly, the zeta potential of NA-Cu@His (15.11 mV) showed a significant decreasing trend, which was explained by the consumption of amino groups during the reaction of acryloyl chloride with Cu@His and the partial coverage of copper ions by the grafted acrylamide group. However, the presence of PEI elevated the zeta potential of PNA-Cu@His to 37.10 mV. Fig. 1d–f show the basic morphologies of the particles. Cu@His are small round particles, whereas NA-Cu@His and PNA-Cu@His become irregular and aggregates into larger particles, which is consistent with the results of particle size tests. Elemental mapping of Cu@His (Fig. 1g) demonstrates the uniform distribution of Cu, C, O, and N elements in the particles.

Changes in the surface chemical state of the prepared particles were further probed by x-ray photoelectron spectroscopy (XPS). XPS mapping of Cu@His (Fig. S1) showed major elements including Cu, C, O and N, confirming the effective binding of histidine to copper ions. In the spectrum of C 1s (Fig. 2a), histidine and Cu@His were categorized into three peaks at 284.79, 285.79, and 288.06 eV, corresponding to C–C/C=C, C–N, and O–C=O (carboxyl), respectively [37,38]. A new peak located at 288.98 eV in NA-Cu@His is identified as C=O (amide). The characteristic peaks in the high-resolution N 1s spectrum of histidine (Fig. 2b) located at 398.38 and 400.69 eV are attributed to the C=N–C and C–NH–C of the imidazole groups, and the peak at 399.79 eV corresponds to the C–NH₂ [39]. It is noteworthy that the peak intensity of the imidazole in Cu@His is remarkably lower, which may imply that the imidazole ring is more likely to form coordination bonds with Cu^{2+} leading to a decrease in the number of imidazole groups on the surface. For NA-Cu@His and PNA-Cu@His, the additional peaks (401.08 and 401.11 eV) belong to the O=C–N in the amide [40]. In Fig. 2c, the O 1s spectra of histidine and Cu@His were deconvoluted into the classical C=O and C–O peaks. The new peak (531.63 eV) that appears in NA-Cu@His is assigned to the C=O bond in the amide. The intensity of the C=O (carboxyl) of PNA-Cu@His was decreased and that of the C=O

(amide) was significantly enhanced after PEI modification, which proved that NA-Cu@His was bound to PEI through amide bonding. In the Cu 2p spectra (Fig. 2d), all curves were fitted to four strong peaks of $\text{Cu}^+ 2p_{1/2}$, $\text{Cu}^{2+} 2p_{1/2}$, $\text{Cu}^+ 2p_{3/2}$, and $\text{Cu}^{2+} 2p_{3/2}$. The strong peak of Cu^+ in Cu@His signifies that a large amount of Cu^{2+} is reduced to Cu^+ during the hydrothermal process. Cu signal of PNA-Cu@His was attenuated, confirming the combination of PEI cladding layer.

The morphologies of the hydrogel coatings after swelling equilibrium in ASW (pH = 7.8) were obtained by scanning electron microscopy, as shown in Fig. 3. It can be observed in Fig. 3a that the PAA hydrogel swelled to form pores with pore diameters of about 7–12 μm . In Fig. 3b, the addition of PEI increased the pore size (5–18 μm) and uneven pore distribution of hydrogel. However, PANA-Cu@His hydrogels show different microscopic morphologies. As can be seen from Fig. 3c–e, the PANA-Cu@His hydrogels formed uniform honeycomb pores with a distinct layer structure. The coating structure becomes denser as the content of PNA-Cu@His increases. The magnified images (Fig. 3f–h) show that the pores of the composite coating are tightly linked by a large number of spiderweb-like microstructures, which are related to the formation of high-density hydrogen bonds between the hydrogel molecules. The dense pore structure provides the basis for the good swelling resistance and mechanical properties of the hydrogels.

3.2. Mechanical and swelling properties of hydrogels

Different hydrogels were prepared as standard sized strips stretched at a uniform speed of 100 mm/min to obtain the tensile curves, and the results are shown in Fig. 4a–b and Fig. S2a. The PANA-Cu@His-1.0 wt% hydrogel exhibits a tensile strength of 3.018 MPa and an elongation at break of 230.39 %, substantially outperforming other PAA-based hydrogels in mechanical properties [41,42]. Moreover, the PANA-Cu@His-1.0 wt% hydrogel does not break even when suspended from 1 kg weights (Fig. 4c). This is due to the complex multilayered cross-linking structure created within the composite hydrogel. Firstly, NA-Cu@His connected the PAA framework and PEI molecular chains

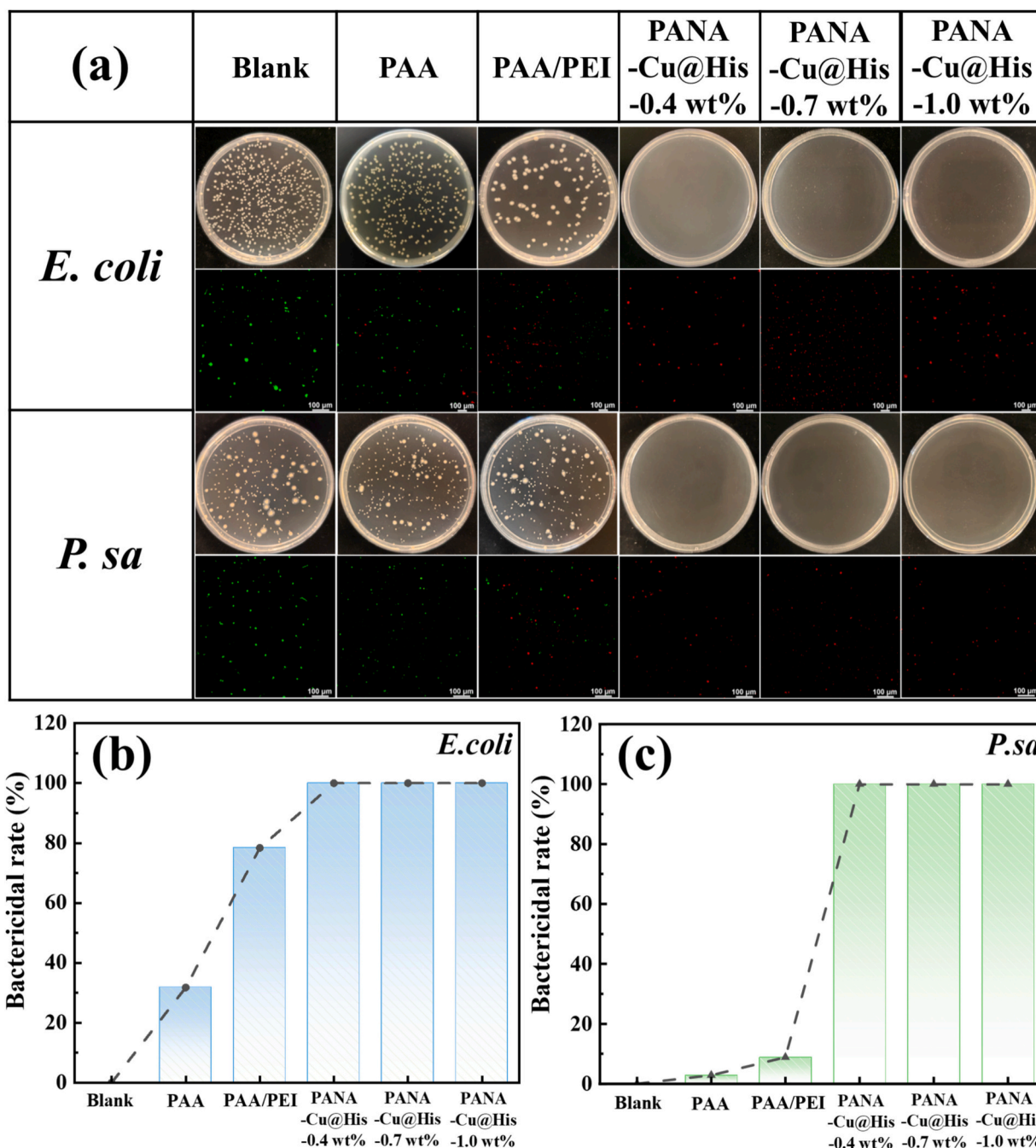


Fig. 6. Images of agar plates and fluorescence microscopy in *E. coli* and *P. sa* after treatment with different hydrogel samples (a); bactericidal rates of different samples against *E. coli* (b) and *P. sa* (c).

through double and amide bonds respectively. Secondly, the coordination between histidine and Cu^{2+} not only increases the intermolecular force, but may also form a dynamic cross-linking network. In addition, the amino groups, carboxyl groups and amide bonds in the composite hydrogel can easily form a large number of intermolecular hydrogen bonds, which also contributes to the improvement of mechanical properties. Excellent mechanical properties are the result of the synergistic action of multiple forces and cannot be achieved by hydrogen bonding (PAA/PEI) or metal coordination (PAA-Cu@His) alone.

The hydrogels were immersed in ASW for 7 days to test the swelling behavior (Fig. 4d) and swelling rate (Fig. 4e and Fig. S2b). The swelling rates of PAA, PAA/PEI and PAA-Cu@His hydrogels were even as high as 174.56 % due to the loose internal network, while the swelling rates of PANA-Cu@His hydrogels were all maintained at less than 100 %. PANA-

Cu@His has a variety of chemically bonded cross-linking structures and high-density hydrogen bonding between the molecular chains, which endows the composite hydrogel with excellent anti-swelling properties. Fig. 4f and Fig. S3 exhibit the adhesion stability of different hydrogel samples after a period of immersion in ASW and real seawater. The adhesion strengths of PAA and PAA/PEI hydrogels were 67 and 75 kPa before immersion and 9 and 14 kPa after immersion, with a decrease in adhesion performance of about 80 %. This is attributed to the loose internal structure of the hydrogel, which causes serious swelling phenomenon due to the large influx of seawater during immersion, resulting in the easy peeling of PAA and PAA/PEI from the substrate (Fig. S3c). In contrast, the PANA-Cu@His hydrogels exhibited superior adhesion strength prior to immersion (109–116 kPa), with a modest reduction of 16–21 % in adhesion strength after immersion, and maintained adhesion

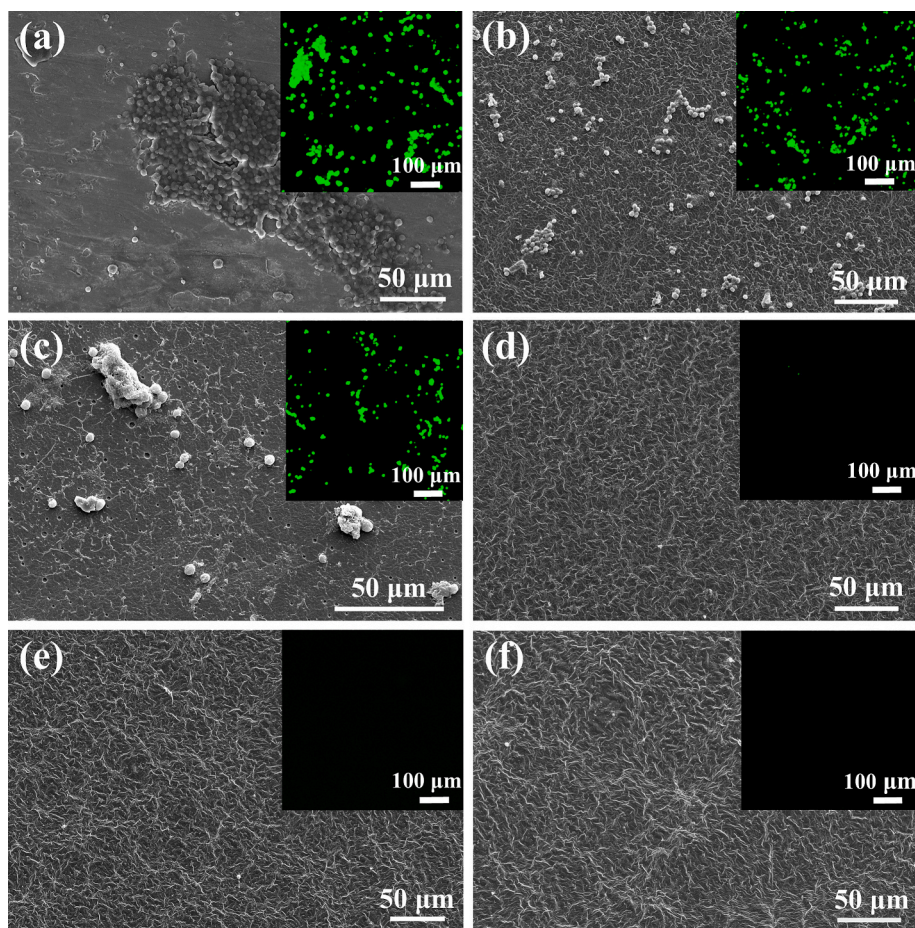


Fig. 7. Anti-*Chlorella* adherence results of titanium alloy (a), PAA (b), PAA/PEI (c), PANA-Cu@His-0.4 wt% (d), PANA-Cu@His-0.7 wt% (e), and PANA-Cu@His-1.0 wt% (f) hydrogels.

stability after 10 days of immersion in real seawater (**Fig. S3d**), which was attributed to the chelation of carboxylate groups with titanium as well as to the high-density cross-linking structure of the composite hydrogel. Enhanced hydrophilic properties facilitate the establishment of a robust hydration layer, thereby markedly augmenting the anti-fouling efficacy of the coating. The water contact angle of the composite hydrogels gradually decreased to 20.8° with increasing PANA-Cu@His content (**Fig. 4g**). In addition to the increase in hydrophilic PEI and amide bonds, it was shown that strong cation- π interaction between histidine and Cu^{2+} significantly enhances the hydrophilicity of histidine [43].

3.3. Acid-sensitive release of Cu^{2+} from composite hydrogels

Marine organisms such as mussels and bacteria will adhere to the substrate and secrete acidic substances, so that the pH value of their attachment site is adjusted to 2.5–6 [44,45]. In order to improve the utilization of antifouling agents, hydrogel coatings with acid-sensitive stimulation of Cu^{2+} release have been developed. The Cu^{2+} leaching concentration (**Fig. 5a**) and the average release rate of Cu^{2+} (**Fig. S4a**) from PANA-Cu@His-1.0 wt% hydrogel in ASW at different pH values were examined by atomic absorption spectroscopy, respectively. When the pH of ASW was decreased to 4 and 2, the Cu^{2+} concentration increased dramatically (43.68 and 57.33 $\mu\text{g/mL}$), which was 127 times higher than that in a neutral environment, showing a highly pronounced pH sensitivity. The average release rate of Cu^{2+} in ASW at pH = 8 was lower than most reported coatings (**Fig. S4b** and **Table S1**), whereas when the pH was lowered to weakly acidic, the Cu^{2+} release rate increased to $7.32 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{d}^{-1}$, which shifted to the “anti-fouling mode”.

In addition, **Fig. 5b** shows that the color of faded after 7 days of immersion in acidic ASW, while there was no significant change in the hydrogel immersed in normal ASW. When the pH of the external environment is neutral or weakly basic, the imidazole ring is deprotonated, and the lone pairs of electrons on its sp^2 -hybridized imidazolium nitrogen atoms can serve as effective Lewis base sites to form stable ligand bonds with Cu^{2+} (**Fig. 5c**). In contrast, the imidazole ring is protonated in an acidic environment to form a positively charged imidazolium ion, at which point both nitrogen atoms are bonded to hydrogen and can no longer provide lone pairs of electrons for coordination with Cu^{2+} [46]. The disappearance of the imidazole- Cu^{2+} peak located at 1523 cm^{-1} in the in-situ infrared spectra in acidic environments also confirms the pH sensitivity of Cu@His (**Fig. S5**).

3.4. Marine antifouling properties of hydrogels

Classical colony counting and live-dead bacterial staining methods were used to evaluate the antimicrobial properties of hydrogels using Gram-negative bacteria *P. sa* and *E. coli*, which are commonly found in the marine environment, as experimental strains. The green dots and red dots observed under a super-resolution fluorescence microscope are live and dead bacteria, and the sterilization results are shown in **Fig. 6a**. After 24 h of incubation, the number of surviving colonies of *E. coli* and *P. sa* in the blank groups were $551 \times 10^3 \text{ CFU/mL}$ and $282 \times 10^3 \text{ CFU/mL}$, respectively. PAA/PEI hydrogels have certain bactericidal properties with bactericidal rates of 31.76 % and 8.87 % against *E. coli* and *P. sa*, respectively (**Fig. 6b** and **c**). This is attributed to the electrostatic disruption mechanism, where the protonated amine groups of PEI interact with the negatively charged bacterial cell wall, leading to

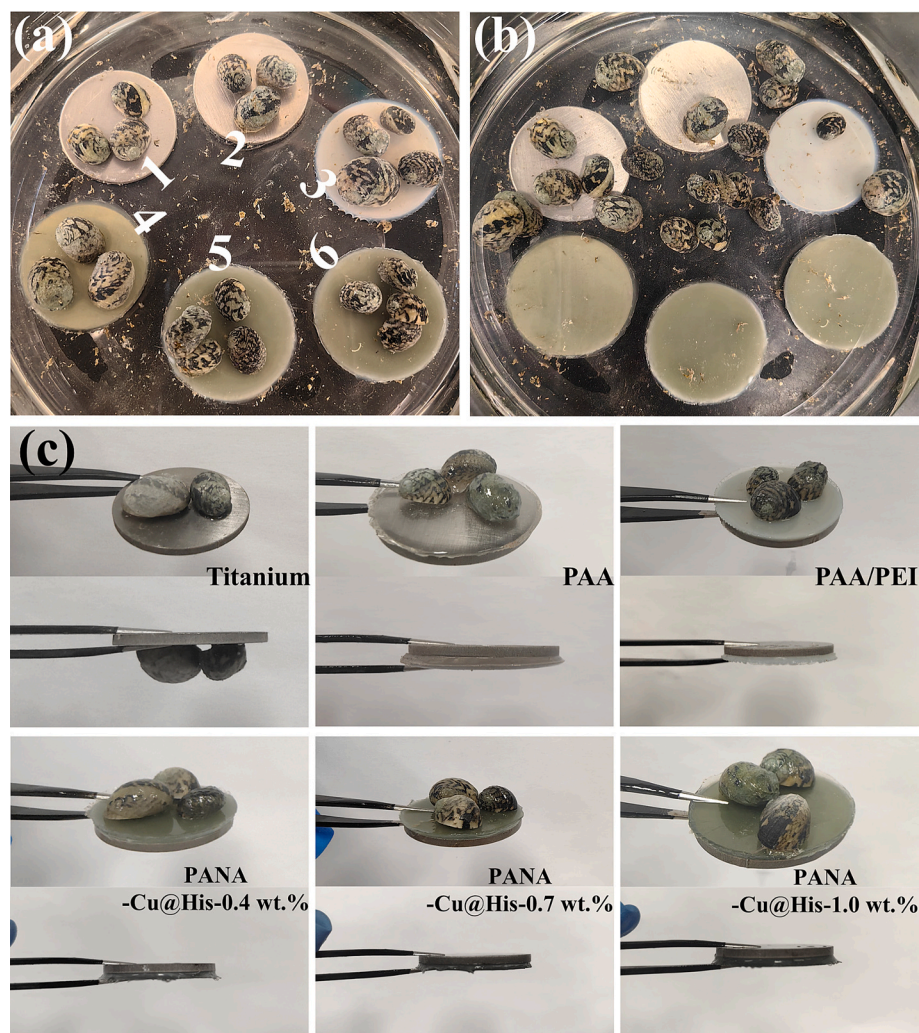


Fig. 8. Pictures of anti-marine organism adhesion test (1: titanium alloy, 2: PAA, 3: PAA/PEI, 4: PANA-Cu@His-0.4 wt%, 5: PANA-Cu@His-0.7 wt%, 6: PANA-Cu@His-1.0 wt%).

membrane depolarization and a limited degree of cell lysis.

Strikingly, no colonies grew on the agar plates of the PANA-Cu@His composite hydrogels, and their corresponding fluorescence images also showed all red dead bacteria, implying that the composite hydrogels have strong bactericidal properties. This remarkable enhancement points to a powerful synergistic “attract-and-kill” bactericidal mechanism. At first, the cationic PEI component aggregates negatively charged bacteria to the hydrogel surface by electrostatic interaction, and this localized enrichment greatly increases the effective concentration of Cu^{2+} delivered to the captured bacteria. Aggregated microbiota form localized acidic zones, triggering a massive release of Cu^{2+} from the coating, which subsequently interferes with cellular metabolism by binding to sulfhydryl groups in proteins and disrupts cell membrane integrity, leading to leakage of intracellular contents and rapid cell death [47].

Resistance to algal adherence is also one of the important indicators of marine antifouling performance of coatings. Images of electron microscopy and laser confocal microscopy give the adherence of *Chlorella* to different hydrogel surfaces. Good biocompatibility makes titanium alloys an ideal habitat for marine organisms, with *Chlorella* congregating in large numbers on the surface of titanium alloys and forming biotopes (Fig. 7a). The PAA and PAA/PEI hydrogels had a large number of sporadic *Chlorella* attached to the surface, but the algae did not aggregate to form a biofilm (Fig. 7b and c). Compared to titanium alloys, PAA and PAA/PEI samples showed improved resistance to algal adherence, which

was attributed to the hydration layer on the hydrogel surface [48]. There was almost no attachment of *Chlorella* to the surfaces of PANA-Cu@His hydrogels after 14 days of continuous incubation in the algal solution, indicating that the composite coatings had excellent resistance to algal adherence. The enhanced hydrophilicity makes it easier to form a hydrated layer on the surface of the composite hydrogels, and the released Cu^{2+} has an inhibitory effect on the growth of *Chlorella* and interferes with the normal metabolic activities of the algae [49].

Besides small fouling organisms such as bacteria and algae, hard-shelled organisms are usually found in the later stages of the fouling phase. The conchs have suction cups and secrete a sticky substance that makes them cling to the substrate (Fig. S6). Natural conchs were placed on different samples and incubated for 3 days to observe the behavior of the conchs. The comparison between Fig. 8a and b shows that the conchs are able to adhere stably to the titanium alloy, while the conchs on the surface of the hydrogel samples have crawled away. Inability of conchs to adhere firmly to hydrogel samples as demonstrated by inversion experiments (Fig. 8c). The hydration layer makes it difficult for the suction cups of the conch to directly contact the surface of the hydrogel, reducing the effective area of attachment. In addition, Cu^{2+} released from PANA-Cu@His hydrogels can inhibit mineralization in conchs, and affect the hardening and deposition of gelatinous substances secreted by marine organisms.

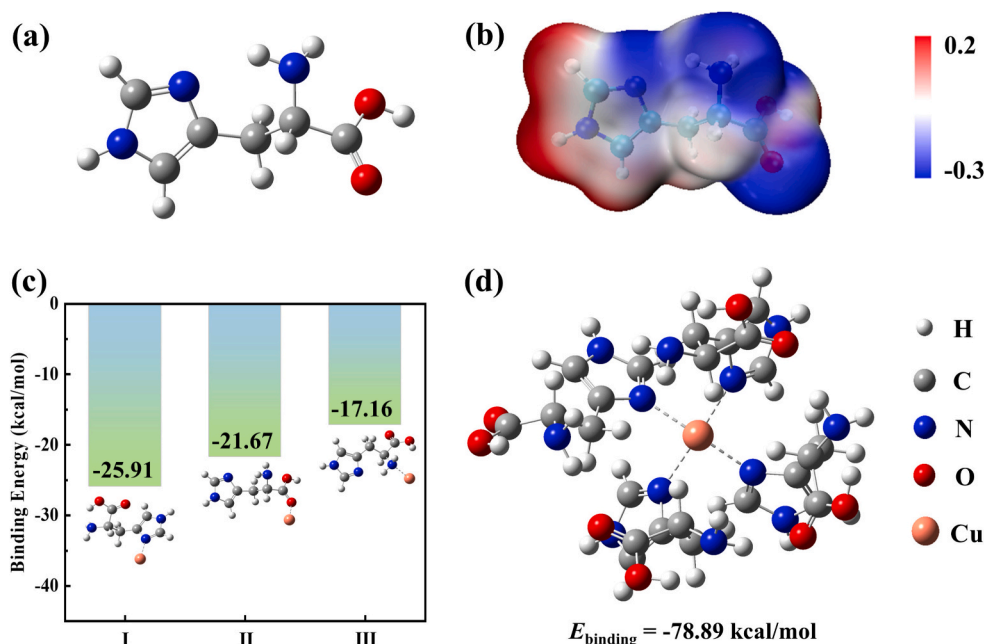


Fig. 9. Molecular structure (a) and electrostatic potential map (b) of histidine; binding energies of Cu^{2+} at different coordination sites on histidine: pyridine N (I), carboxyl (II), amino (III) (c); the configuration of Cu@His calculated by DFT (d).

3.5. DFT calculation

Fig. 9a depicts the optimized molecular configuration of histidine via Gaussian 16. The four potential binding sites for Cu^{2+} in L-histidine are the carboxyl group, the amino group, the pyrrole N in imidazole, and the pyridine N. While all four positions possess coordination capability via their lone pair electrons, the pyrrole nitrogen exhibits significantly diminished ligand strength due to its participation in the aromatic π -system. As illustrated in Fig. 9b, the positive electrostatic potential at the pyrrole nitrogen of imidazole ring indicates its electron deficiency, thereby precluding its participation in Cu^{2+} coordination. To elucidate the preferential coordination of Cu^{2+} within the histidine, the binding energies of different configurations were systematically evaluated through DFT calculations incorporating dispersion-corrected functionals. Among the investigated coordination modes, the pyridine nitrogen-coordinated configuration exhibited exceptional thermodynamic stability, with a calculated binding energy of -25.91 kcal/mol, thereby establishing this as the predominant Cu^{2+} coordination site (Fig. 9c). DFT calculations reveal that copper is primarily coordinated by the imidazole rings, and the optimal configuration is four imidazole rings forming coordination bonds with Cu^{2+} , as shown in Fig. 9d. This “imidazole-dominated and carboxyl-assisted” coordination unit creates local pH-responsive domains that accelerate Cu^{2+} release only when microbial colonization acidifies the microenvironment, enabling “on-demand” antifouling that reduces copper loading compared to conventional cuprous oxide coatings while maintaining equivalent performance against major fouling species.

4. Conclusion

In conclusion, we prepared an acid-sensitive antifouling hydrogel coating by utilizing the pH sensitivity and coordination ability of histidine. DFT simulations show that the binding energies of imidazolyl- Cu^{2+} and carboxyl- Cu^{2+} are -25.91 and -21.67 kcal/mol, which confirms the “imidazole-dominated and carboxyl-assisted” coordination configuration of Cu@His. The localized acidic environment formed by the attachment of fouling organisms can stimulate PANA-Cu@His hydrogel to release 127 times more Cu^{2+} antifouling agents than in a neutral environment, achieving strong bactericidal effect and anti-algae

adhesion performance. The integration of multifaceted hydrogen-bonding networks within hydrogel matrices enhanced the mechanical robustness of coatings by achieving an elongation at break and tensile strength of 230.39 % and 3.018 MPa. This pH-responsive antifouling hydrogel coating achieves precise control over biocide release while effectively mitigating heavy metal leaching into marine ecosystems, thereby enhancing the environmental sustainability of copper-based antifouling strategies in marine engineering applications.

CRediT authorship contribution statement

Qianyun Tan: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Gege Zhang:** Investigation, Data curation. **Xiaoyu Xu:** Investigation, Data curation. **Tiantian Hu:** Investigation. **Yabei Zhang:** Supervision. **Xu Zhang:** Data curation. **Zhongyao Zhang:** Software, Methodology, Formal analysis, Data curation. **Fa-Qian Liu:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

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Declaration of competing interest

The authors declare no competing interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.susmat.2025.e01521>.

Data availability

Data will be made available on request.

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